BCS401

Model Question Paper with effect from 2023-24 (CBCS Scheme)

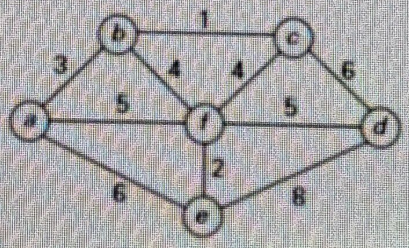
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Fourth Semester B.E. Degree Examination

Analysis and Designs of Algorithms

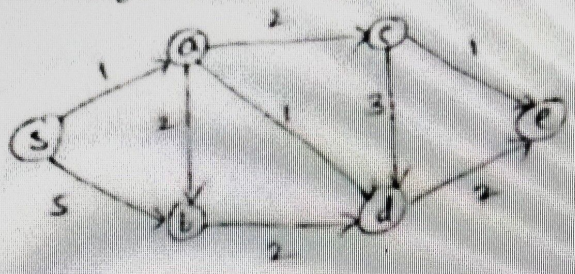
TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

		Module -1	BL		Marks
Q.01	a	Define algorithm. Explain asymptotic notations Big Oh, Big Omega and Big Theta notations	L2		08
	b	Explain the general plan for analyzing the efficiency of a recursive algorithm. Suggest a recursive algorithm to find factorial of number. Derive its efficiency	L2		08
	c	If $t1(n) \in O(g1(n))$ and $t2(n) \in O(g2(n))$, then show that $t1(n) + t2(n) \in O(\max\{g1(n), g2(n)\})$.	L2		04
		OR			
Q.02	a	With neat diagram explain different steps in designing and analyzing an algorithm	L2		08
	b	Explain the general plan for analyzing the efficiency of a non-recursive algorithm. Suggest a non-recursive algorithm to find maximum element in the list of n numbers. Derive its efficiency	L2		08
	c	With the algorithm derive the worst case efficiency for Bubble sort	L2		04
		Module-2			
Q. 03	a	Explain the concept of divide and conquer. Design an algorithm for merge sort and derive its time complexity	L2		10
	b	Design an insertion sort algorithm and obtain its time complexity. Apply insertion sort on these elements. 25,75,40,10,20.	L3		10
		OR			
Q.04	a	Explain Strassen's matrix multiplication and derive its time complexity	L2		10
	b	Design an algorithm for quick sort algorithm. Apply quick sort on these elements. 25,75,40,10,20,05,15	L3		10
		Module-3			
Q. 05	a	Define AVL Trees. Explain its four rotation types	L2		10
	b	Construct bottom up heap for the list 2,9,7,6,5,8. Obtain its time complexity	L3		10
		OR			
Q. 06	a	Define heap. Explain the properties of heap along with its representation.	L2		10
	b	Design Horspools algorithm for string matching. Apply Horspools algorithm to find the pattern BARBER in the text: JIM_SAW_ME_IN_A_BARBERSHOP	L3		10
		Module-4			
Q. 07	a	Construct minimum cost spanning tree using Kruskals algorithm for the following graph. 	L3		10
	b	What are Huffman Trees? Construct the Huffman tree for the following data.	L3		10

Character	A	B	C	D	E	-
Probability	0.5	0.35	0.5	0.1	0.4	0.2

Encode DAD-CBE using Huffman Encoding.

		OR			
Q. 08	a	Apply Dijkstra's algorithm to find single source shortest path for the given graph by considering S as the source vertex.  	L3		10
	b	Define transitive closure of a graph. Apply Warshalls algorithm to compute transitive closure of a directed graph $\begin{array}{c} \\ a \\ b \\ c \\ d \end{array}\begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$	L3		10
		Module-5			
Q. 09	a	Explain the following with examples i) P problem ii) NP Problem iii) NP- Complete problem iv) NP – Hard Problems	L2		10
	b	What is backtracking? Apply backtracking to solve the below instance of sum of subset problem S={5,10,12,13,15,18} d=30	L3		10
		OR			
Q. 10	a	Illustrate N queen's problem using backtracking to solve 4-Queens problem	L2		10
	b	Using Branch and Bound technique solve the below instance of knapsack problem. Capacity 5	L3		10

Item	Weight	Value
1	2	12
2	1	10
3	3	20
4	2	5

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BCS401

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Analysis and Design of Algorithms

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

			M	L	C
		Module – 1			
Q.1	**a.**	What is an algorithm? Explain the fundamentals of algorithmic problem solving.	**10**	**L2**	**CO1**
	b.	Develop an algorithm to search an element in an array using sequential search. Calculate the best case, worst case and average case efficiency of this algorithm.	**10**	**L3**	**CO1**
		OR			
Q.2	**a.**	Explain asymptotic notations with example.	**10**	**L2**	**CO1**
	b.	Give the general plan for analyzing the efficiency of the recursive algorithm. Develop recursive algorithm for computing factorial of a positive number. Calculate the efficiency in terms of order of growth.	**10**	**L3**	**CO1**
		Module – 2			
Q.3	**a.**	Explain Strassen's matrix multiplication approach with example and derive its time complexity.	**10**	**L3**	**CO2**
	b.	What is divide and conquer? Develop the quick sort algorithm and write its best case. Make use of this algorithm to sort the list of characters: E, X, A, M, P, L, E.	**10**	**L2**	**CO2**
		OR			
Q.4	**a.**	Distinguish between decrease & conquer and divide & conquer algorithm design techniques with block diagram. Develop insertion sort algorithm to sort a list of integers and estimate the efficiency.	**10**	**L3**	**CO2**
	b.	Define topological sorting. List the two approaches of topological sorting and illustrate with examples.	**10**	**L2**	**CO2**
		Module – 3			
Q.5	**a.**	Define AVL tree with an example. Give worst case efficiency of operations on AVL tree. Construct an AVL tree of the list of keys: 5, 6, 8, 3, 2, 4, 7 indicating each step of key insertion and rotation.	**10**	**L3**	**CO3**
	b.	Define Heap. Explain the bottom-up heap construction algorithm. Apply heap sort to sort the list of numbers 2, 9, 7, 6, 5, 8 in ascending order using array representation.	**10**	**L3**	**CO3**
		OR			
Q.6	**a.**	Define 2-3 tree. Give the worst case efficiency of operations on 2-3 tree. Build 2-3 tree for the list of keys 9, 5, 8, 3, 2, 4, 7 by indicating each step of key insertion and node splits.	**10**	**L3**	**CO3**
	b.	Design Horspool algorithm for string matching. Apply this algorithm to find the pattern BARBER in the text: JIM_SAW_ME_IN_A_BARBERSHOP	**10**	**L3**	**CO3**
		Module – 4			
Q.7	**a.**	Apply Dijkstra's algorithm to find the single source shortest path for given graph [Fig.Q7(a)] by considering 's' as source vertex. Illustrate each step.	**10**	**L3**	**CO4**

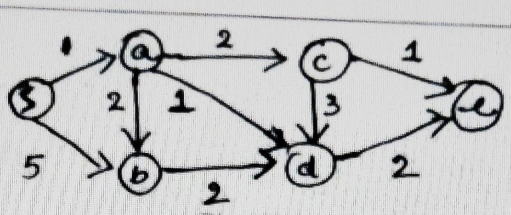


Fig.Q7(a)

	b.	Define transitive closure. Write Warshall's algorithm to compute transitive closure. Illustrate using the following directed graph.	10	L3	CO4

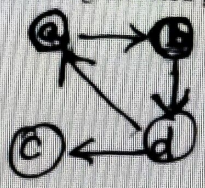


Fig.Q7(b)

OR

Q.8	a.	Define minimum spanning tree. Write Kruskal's algorithm to find minimum spanning tree. Illustrate with the following undirected graph.	10	L3	CO4

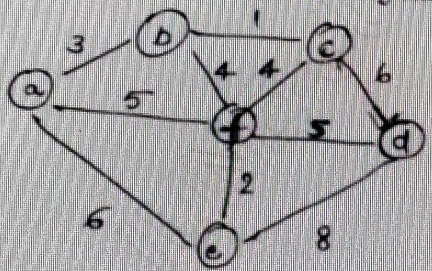


Fig.Q8(a)

	b.	Construct Huffman Tree and resulting code for the following: (i) Encode the text : ABACABAD (ii) Decode the text : 100010111001010	10	L3	CO4

Character	A	B	C	D	-
Probability	0.4	0.1	0.2	0.15	0.15

Module – 5

Q.9	a.	Explain n-Queen's problem with example using backtracking approach.	10	L2	CO5
	b.	Solve the following instance of the knapsack problem by the branch-and-bound algorithm. Construct state-space tree.	10	L3	CO5

Item	Weight	Value
1	4	$ 40
2	7	$ 42
3	5	$ 25
4	3	$ 12

The knapsack's capacity W is 10.

OR

Q.10	a.	Differentiate between Branch and Bound technique and Backtracking. Apply backtracking to solve the following instance of subset-sum problem S = {3, 5, 6, 7} and d = 15. Construct a state space tree.	10	L3	CO5
	b.	Explain greedy approximation algorithm to solve discrete knapsack problem.	10	L2	CO5

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Modified

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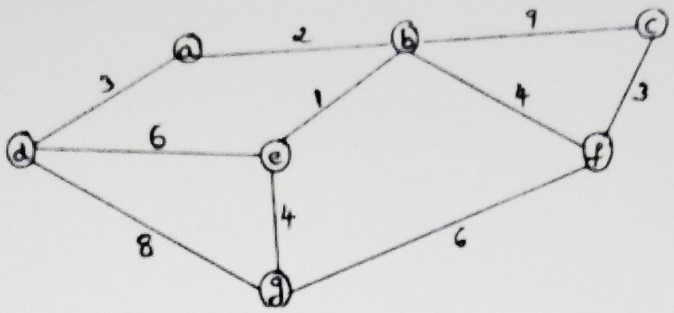
Fourth Semester B.E./B.Tech. Degree Examination, June/July 2025
Analysis and Design of Algorithms

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

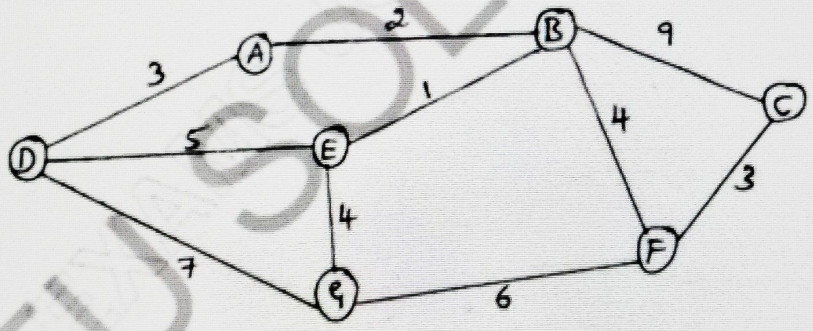
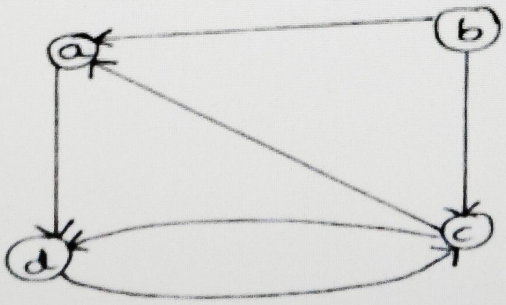
			M	L	C
		Module – 1			
Q.1	**a.**	Define algorithm Explain asymptotic notations Bigh oh, Big omega and Big theta notations.	**08**	**L2**	**CO1**
	b.	Explain the general plan for analyzing the efficiency of a recursive algorithm. Suggest a recursive algorithm to find factorial of number. Derive its efficiency.	**08**	**L3**	**CO1**
	c.	If $t_1(n) \in O(g_1(n))$ and $t_2(n) \in O(g_2(n))$ then show that $t_1(n) + t_2(n) \in o(\max\{g_1(n), g_2(n)\})$	**04**	**L2**	**CO1**
		OR			
Q.2	**a.**	With a neat diagram explain different steps in designing and analyzing algorithm.	**08**	**L2**	**CO1**
	b.	Write an algorithm to find the max element in an array of n elements. Give the mathematical analysis of this non- recursive algorithm.	**08**	**L3**	**CO1**
	c.	With the algorithm derive the worst case efficiency for selection sort.	**04**	**L3**	**CO1**
		Module – 2			
Q.3	**a.**	Explain the concept of divide and conquer. Design an algorithm for merge sort and derive its time complexity.	**10**	**L3**	**CO2**
	b.	Design an algorithm for insertion algorithm and obtain its time complexity. Apply insertion sort on these elements. 89, 45, 68, 90, 29, 34, 17	**10**	**L3**	**CO2**
		OR			
Q.4	**a.**	Design an algorithm for Quick sort. Apply quick sort on these elements. 5, 3, 1, 9, 8, 2, 4, 7.	**10**	**L3**	**CO2**
	b.	Explain Strassen's Matrix multiplication and derive its time complexity.	**10**	**L2**	**CO2**
		Module – 3			
Q.5	**a.**	Define AVL trees. Explain its four rotation types.	**10**	**L2**	**CO3**
	b.	Design an algorithm for Heap sort. Construct bottom – up heap for the list 15, 19, 10, 7, 17, 16.	**10**	**L3**	**CO4**
		OR			
Q.6	**a.**	Design Horspool's Algorithm for string matching Apply Horspool algorithm to find pattern BARBER in the test: JIM_SAW_ME_IN_A_BARBERSHOP.	**10**	**L3**	**CO4**
	b.	Define heap. Explain the properties of heap along with its representation.	**10**	**L2**	**CO3**

Module – 4

Q.7	**a.**	Construct minimum cost spanning tree using Kruskal's algorithm for the following graph.  Fig. 7(a)	**10**	**L3**	**CO4**
	b.	What are Huffman trees? Construct the Huffman tree for the following data i) Encode the text ABAC ABAD ii) Decode the code100010111001010	**10**	**L3**	**CO4**

Character	A	B	C	D	-
Probability	0.4	0.1	0.2	0.15	0.15

OR

Q.8	**a.**	Apply Dijkstra's algorithm to fine single source shortest path for the given graph by considering A as the source vertex.  Fig.8 (a)	**10**	**L3**	**CO4**
	b.	Define transitive closure of a graph. Apply Warshall's algorithm to compute transitive closure of a directed graph.  Fig.8 (b)	**10**	**L3**	**CO4**

VTUSOLUTIONS

Module – 5

Q.9	**a.**	Explain the following with examples. i) P problem ii) NP problem ii) NP-Complete problem iv) NP – Hard problem	**10**	**L2**	**CO5**
	b.	What is backtracking? Apply backtracking to solve the below instance of sum of subset problem. S = { 1, 2, 5, 6, 8} and d = 9.	**10**	**L3**	**CO6**
		OR			
Q.10	**a.**	Illustrate N Queen's problem using backtracking to solve 4 – Queens problem.	**10**	**L2**	**CO6**
	b.	Using Branch and Bound method solve the below instance of Knapsack Problem.	**10**	**L3**	**CO6**

Item	Weight	Value
1	4	40
2	7	42
3	5	25
4	3	12

Capacity = 10

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VTU SOLUTIONS